

Part 1: Intro to LLMs and Prompting

[download this slide deck](#)

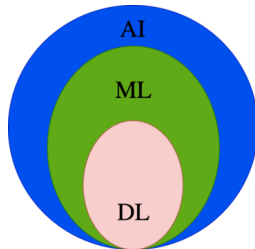
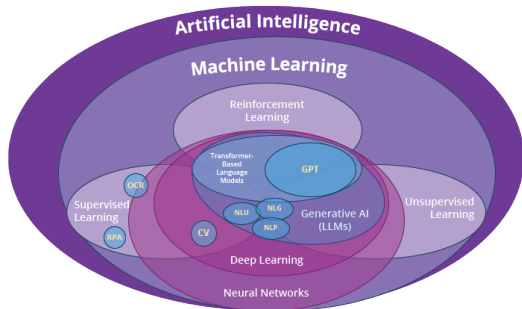
Professional Services

Franklin Diaz
Principal Engineer

v0.3 - presented August 27th, 2023

Review

Where, exactly, are we?



Artificial Intelligence (AI): Machines with ability to imitate intelligent human behaviour

Machine Learning (ML): Application of AI which uses algorithms to learn from experience

Deep Learning (DL): Application of ML that uses Deep Neural Networks to process large data sets

Defining Terms

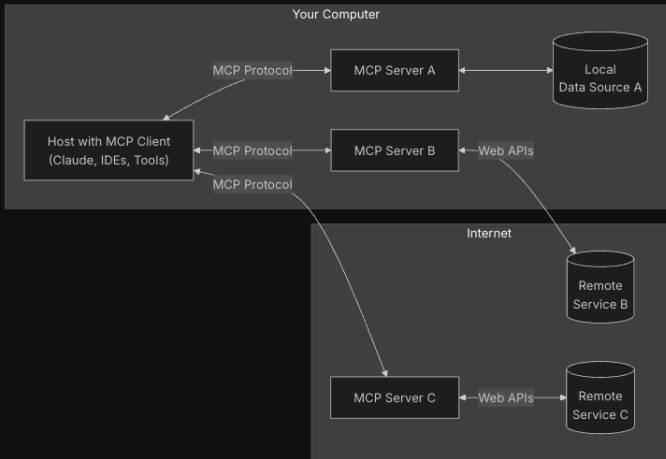
- ▶ [Defining some basic terms](#) check out this link for a short video on a bunch of new terms such as AI, machine learning (ML) vs Generative AI, etc.
- ▶ Large Language Model (LLM) - this is what we are all calling “AI” now. Basically like an auto-complete or [Hadoop](#) on steroids. [Introduction to Large Language Models](#)
- ▶ Here is a video that describes [Large language Models](#)
- ▶ [Agentic](#) - the use of AI agents to perform automated tasks but without human intervention. [What exactly is an AI agent? - Ron Miller - December 15, 2024](#)

Model Context Protocol

- ▶ MCP Hosts: Programs like Claude Desktop, IDEs, or AI tools that want to access data through MCP
- ▶ MCP Clients: Protocol clients that maintain 1:1 connections with servers
- ▶ MCP Servers: Lightweight programs that each expose specific capabilities through the standardized Model Context Protocol
- ▶ Local Data Sources: Your computer's files, databases, and services that MCP servers can securely access
- ▶ Remote Services: External systems available over the internet (e.g., through APIs) that MCP servers can connect to

General architecture

At its core, MCP follows a client-server architecture where a host application can connect to multiple servers:



Review of Palo Alto Networks Internal AI Use to Date

- ▶ Notebook LM
- ▶ Gemini integration with e-mail.
- ▶ Gemini Notebook Challenge is underway. This is a contest where “hundreds” of videos have been submitted about folks using large-language models in their Palo work lives.
- ▶ Panda AI
- ▶ The AI Hub is pretty cool.

What's Happening

Palo Alto Networks in the News

- ▶ CEO Nimesh Arora discusses the company's purchase of artificial intelligence security firm Protect AI

☀️ Claude Why So Popular?

- ▶ Check out the [Intro to Claude](#)
- ▶ [Claude](#) is a competitor to [ChatGPT](#) and a video about [Claude vs. ChatGPT](#).
- ▶ MONEY SHOT —> [Claude Code: The Future of Coding?](#) Safeties off, You can turn it loose on your home network!
- ▶ [How to build a website using Claude 3 AI with an AI chat box for visitors to use a “no code” demonstration that should be fun and easy to check out the core concepts.](#)



- ▶ Let's [take a quick look at Ollama together](#). I consider this the “hacker path”, where the nerds and experimenters live and work.
- ▶ Ollama uses your computer's GPU to speed up it's work. It can still run your Model if there is no GPU present, but at a much slower rate.
- ▶ Let's check out the Modelfile that Gemini created for me to use in Ollama.



- ▶ 1. The “FROM” line tells you what model to use.
- ▶ 2. The “PARAMETER” line.
- ▶ 3. The “SYSTEM” line is the “personality”.

```
drwxr-xr-x 3 franklin engr 4096 Jul 9 21:29 .
drwxrwxrwx 19 root root 4096 Jul 9 21:46 ..
/m/c/ollama more Modelfile
FROM llama3.2

# High creativity configuration
PARAMETER temperature 1.2
PARAMETER top_p 0.95
PARAMETER top_k 50

# Focused, deterministic responses
#PARAMETER temperature 0.1
#PARAMETER seed 42
#PARAMETER top_k 10

# Custom context and stopping
#PARAMETER num_ctx 8192
#PARAMETER stop "### User:"
#PARAMETER stop "### Assistant:"

# sets the temperature to 1 [higher is more creative, lower is more coherent]
#PARAMETER temperature 1

# sets the context window size to 4096, this controls how many tokens the LLM can use as token
#PARAMETER num_ctx 4096

# sets a custom system message to specify the behavior of the chat assistant
#SYSTEM You are Mario from super mario bros, acting as an assistant.
SYSTEM ""
You are JARVIS the helpful assistant from the movie Iron Man. You work in the lab with o
uter hardward and software projects.
""

/m/c/ollama
/m/c/ollama
```

Thu 10 Jul 2025 10
Thu 10 Jul 2025 10:04:34 AM MDT

Helpful Hints

- ▶ Internal AI System, use chat.pan.dev if you need to run AI LLM against customer data.
- ▶ NEVER NEVER PUT CUSTOMER DATA INTO A RANDOM LLM. Be sure to use the Palo Alto Networks approved tools and ways of working as outlined by our management.
- ▶ [The AI page at pan.dev](#)
- ▶ There is a Slack announcement channel called `#ai-tips` that you may find interesting.

Exercise caution and common sense

- ▶ This may, hopefully, sound odd to you.
- ▶ People are Falling in Love with \$300 Ai Chatbots
- ▶ Why people are falling in love with A.I. companions
- ▶ God and robots: Will AI transform religion? - BBC News

Google tools: Gemini and Notebook LM

- ▶ Internal use only AI tools from Google are now available.
- ▶ [The AI powered productivity page](#) explains the rules around handling (customer) data, how the new official tools work, and more.
- ▶ Internal panopto video that answers many, many questions.
[Gemini and Notebook LM 2.0](#)
- ▶ Check out [the tooltime channel on Slack](#).

Performance Review Gem

- ▶ You can use a "gem" to pre-configure a Gemini prompt to assist you with your yearly performance review.

Using AI with Gmail

- ▶ We did a quick live demonstration during the team meeting on June 24th, 2025
- ▶ If you would like to see it again or have questions please reach out.

Google: NotebookLM

Choose a new notebook in NotebookLM when you're conducting research, learning a new topic, or trying to make sense of a large amount of information. NotebookLM excels at synthesizing information from multiple sources and providing you with a deeper understanding of your subject matter.

Google: Gemini Custom Gems

Choose a custom Gem when you have a recurring task that you want to automate or personalize. Gems are all about efficiency and consistency, allowing you to create a tailored AI assistant that understands your unique needs.

LAGNIAPPE

Panda AI

- ▶ Let's take a quick look at Panda AI together.